

Week 6 – Path Planning

1 Introduction

The goal of this assignment was to implement a path planning algorithm capable of navigating a robot from a start position to a goal position in a given map. The algorithm must avoid obstacles, maintain a safe distance from them, and produce a smooth trajectory.

2 Task 1 – Path Planning Algorithm

For path planning, the A* (A-star) algorithm was implemented. The environment is represented as an occupancy grid obtained from `read_only_vars.discrete_map.map`, where: 0 represents free space, 1 represents obstacles.

The A* algorithm computes the optimal path using:

- cost function $g(n)$ (distance from start),
- heuristic $h(n)$ (Euclidean distance to goal).

The algorithm expands nodes in 8 directions (including diagonals) while preventing diagonal corner cutting.

3 Task 2 – Clearance from Obstacles

To ensure a minimum clearance of 0.2m, obstacle inflation was applied. Each obstacle cell was expanded by a radius corresponding to the required clearance. This effectively reduces the free space but guarantees that the resulting path is safe. Additionally, a geometric collision check was performed using the original wall representation. The distance from each trajectory segment to walls was computed, and paths violating the clearance constraint were rejected.

4 Task 3 – Path Smoothing

The raw A* path was further improved using an iterative smoothing algorithm:

$$x_i = x_i + \alpha(x_i^{orig} - x_i) + \beta(x_{i-1} + x_{i+1} - 2x_i) \quad (1)$$

The smoothed path is used only if it remains collision-free. Otherwise, the original A* path is preserved.

5 Trajectory Following

The robot follows the path using a simple waypoint-based controller. The robot continuously moves towards the next waypoint, adjusting its heading using proportional control.

Key improvements:

- reduced oscillations by limiting angular velocity,
- slower speed for large heading errors,
- skipping close waypoints to improve stability.

6 Outdoor Map Problem

During testing, a significant issue appeared in the outdoor map scenario. Unlike the indoor map, the outdoor map does not provide a reliable `mocap_pose`. Instead, only GNSS-based position is available, which is noisy and less accurate. This leads to: mismatch between planned path and actual robot position, deviation from trajectory, unstable tracking behavior.

6.1 Solution Approach

Several strategies were applied:

- initializing the robot pose manually,
- aligning the first waypoint with the robot position,
- simplifying the controller and limiting angular velocity.

However, even with these improvements, the outdoor performance is still less reliable.

Conclusion: The limitation is mainly caused by localization uncertainty, not the path planning algorithm itself.

7 Results

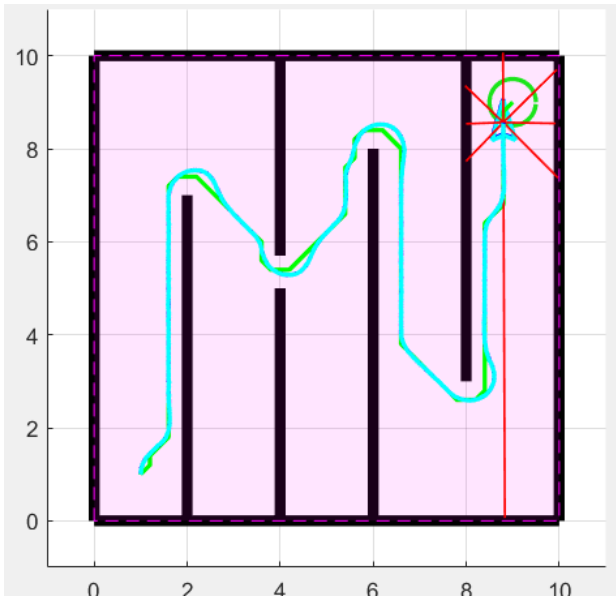


Figure 1: Indoor map trajectory(task 1,2,3)

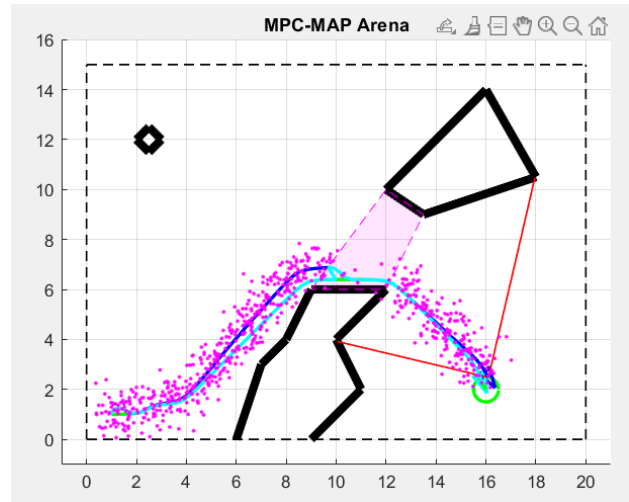


Figure 2: Outdoor map trajectory

8 Discussion

The A* algorithm proved to be reliable and efficient for grid-based path planning. The addition of clearance significantly improved safety. Path smoothing enhanced the visual quality and reduced sharp turns.

The main limitation of the system is not the planner, but the localization accuracy in outdoor environments. Without accurate pose estimation, even a perfect path cannot be followed precisely.

9 Conclusion

A complete path planning pipeline was successfully implemented: A* path planning, obstacle clearance, path smoothing, trajectory following.

The system works reliably in indoor environments. In outdoor scenarios, performance is limited by noisy localization, which should be addressed in future work.